

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1102

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture.
(BL3)

Assessment Tool Weightage: 40%

QUESTIONS: Any 4

Total Marks:40

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I __Pavithra Sri S(192572079)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

ANALYTICAL PROBLEM SOLVING

1. Use Case Diagram Problem

A **Hospital Management System** allows patients to book appointments, doctors to update medical records, and administrators to manage hospital resources. The requirements are not clearly documented.

Tasks:

(a) Identify All Actors Involved in the System

The actors are the users or external entities that interact with the system.

1. **Patient**
2. **Doctor**
3. **Administrator (Admin)**

(b) List the Primary Use Cases

Patient

- Register/Login
- Book Appointment
- View Appointment
- Cancel Appointment

Doctor

- View Appointment Schedule
- Update Medical Records
- Prescribe Medicines

Administrator

- Manage Hospital Resources
- Manage Doctors
- Manage Patients
- Generate Reports

(c) Define Relationships (Include, Extend, Generalization)

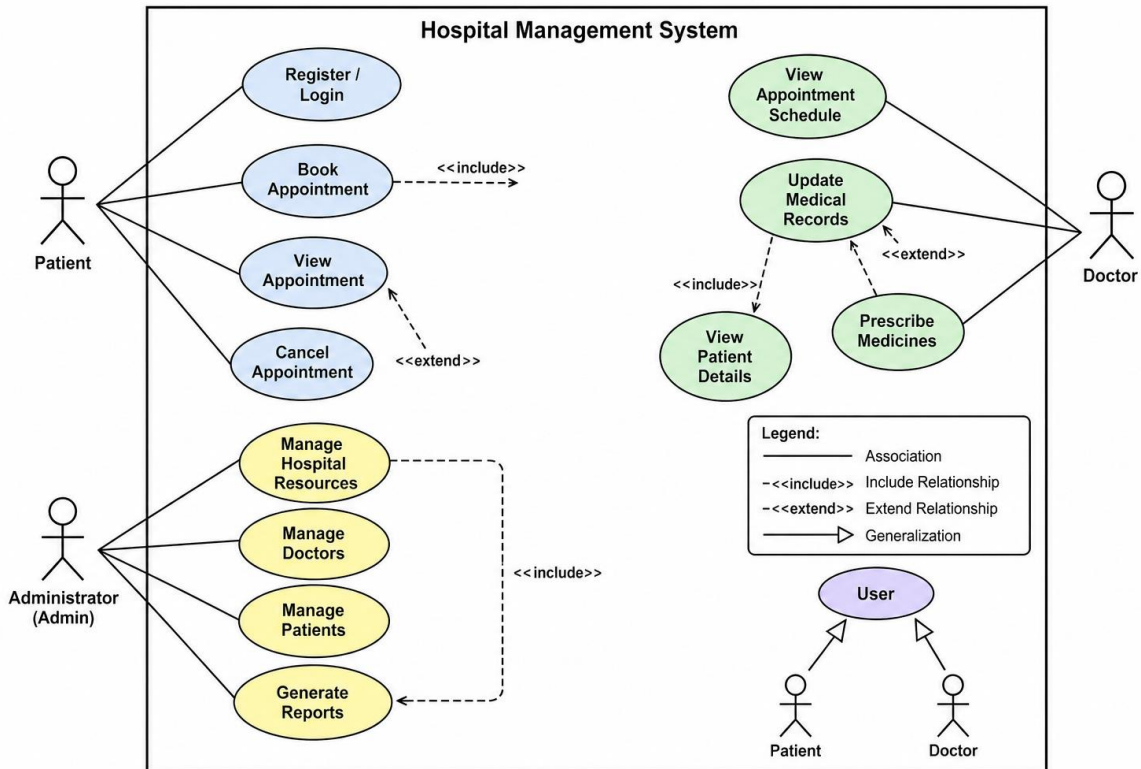
1. Include Relationship (<<include>>)

A mandatory function that is always performed as part of another use case.

Examples:

- **Book Appointment** <<include>> **Login**
- **Update Medical Records** <<include>> **View Patient Details**
- **Manage Hospital Resources** <<include>> **Generate Reports**

(d) Complete Use Case Diagram



(10 Marks)

2. Class Diagram Problem

A **Library Management System** contains entities such as Book, Member, Librarian, and IssueRecord. The relationships among entities are not properly defined.

Tasks:

(a) Identify Classes with Attributes and Methods

1. Book

Attributes:

- BookID
- Title
- Author

- Publisher
- ISBN
- Status

Methods:

- AddBook()
- UpdateBook()
- SearchBook()
- CheckAvailability()

2. Member

Attributes:

- MemberID
- Name
- Address
- PhoneNumber
- Email

Methods:

- Register()
- BorrowBook()
- ReturnBook()
- ViewIssuedBooks()

3. Librarian

Attributes:

- LibrarianID
- Name
- EmployeeID

Methods:

- IssueBook()
- ReceiveBook()
- AddBook()
- RemoveBook()

4. IssueRecord

Attributes:

- IssueID
- IssueDate
- DueDate
- ReturnDate
- FineAmount

Methods:

- CalculateFine()
- UpdateReturnDate()
- GenerateRecord()

(b) Define Relationships

1. Association

Association represents a connection between two independent classes.

- **Member** ↔ **IssueRecord**
- **Book** ↔ **IssueRecord**
- **Librarian** ↔ **IssueRecord**

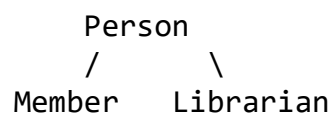
2. Aggregation

Aggregation represents a "**has-a**" relationship where the contained object can exist independently.

- **Library** ◇— **Book**
- A library contains many books, but books can exist independently.

3. Inheritance (Generalization)

Create a common superclass **Person**.

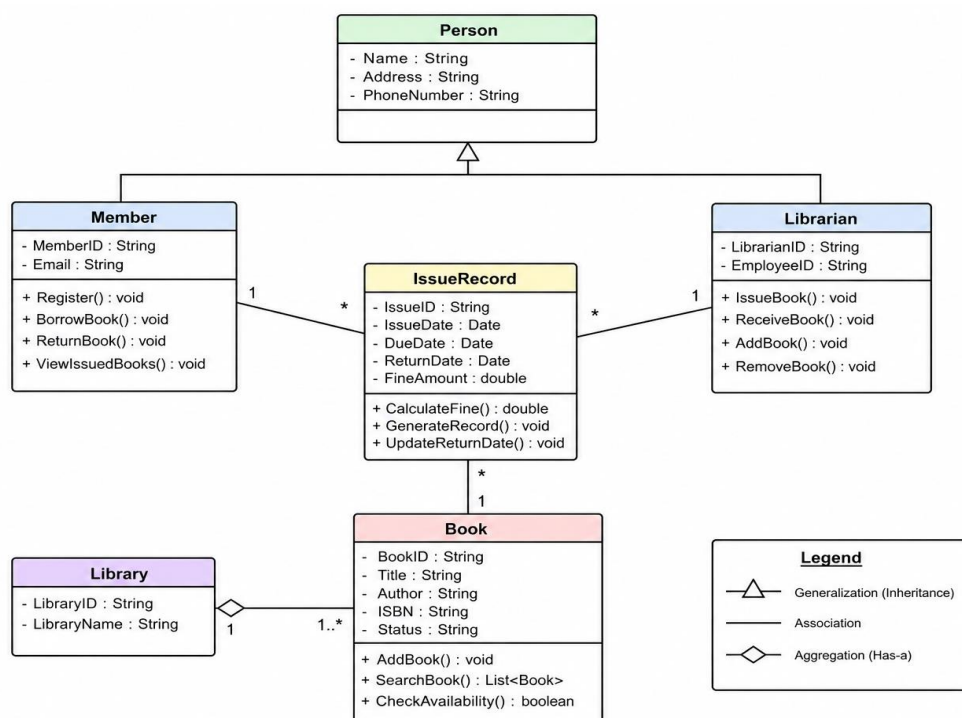


Both **Member** and **Librarian** inherit common attributes such as Name and Address from **Person**.

(c) Specify Multiplicity Between Classes

Classes	Multiplicity	Meaning
Library — Book	1..*	One library contains many books.
Member — IssueRecord	1..*	One member can have many issue records.
Book — IssueRecord	1..*	One book can be issued many times over time.
Librarian — IssueRecord	1..*	One librarian manages many issue records.

(d) Detailed Class Diagram



(10 Marks)

3. Sequence Diagram Problem

In an **Online Food Ordering System**, a customer selects food items, places an order, makes payment, and receives order confirmation. The interaction flow is unclear.

Tasks:

(a) Identify Objects Involved in the Process

The objects (participants) involved in the Online Food Ordering System are:

1. **Customer**
2. **Food Ordering System**
3. **Restaurant**
4. **Payment Gateway**
5. **Order Database**

(b) Define Message Flow Between Objects

The sequence of messages exchanged between the objects is:

1. Customer → Food Ordering System : Select Food Items
2. Customer → Food Ordering System : Place Order
3. Food Ordering System → Restaurant : Send Order Details
4. Restaurant → Food Ordering System : Confirm Order
5. Food Ordering System → Payment Gateway : Process Payment
6. Payment Gateway → Food Ordering System : Payment Successful
7. Food Ordering System → Order Database : Save Order
8. Order Database → Food Ordering System : Order Saved
9. Food Ordering System → Customer : Send Order Confirmation

(c) Represent Activation and Lifelines

Lifelines

Each object has a **vertical dashed line**, representing its existence during the interaction.

Objects with Lifelines:

- Customer
- Food Ordering System
- Restaurant
- Payment Gateway
- Order Database

Activation

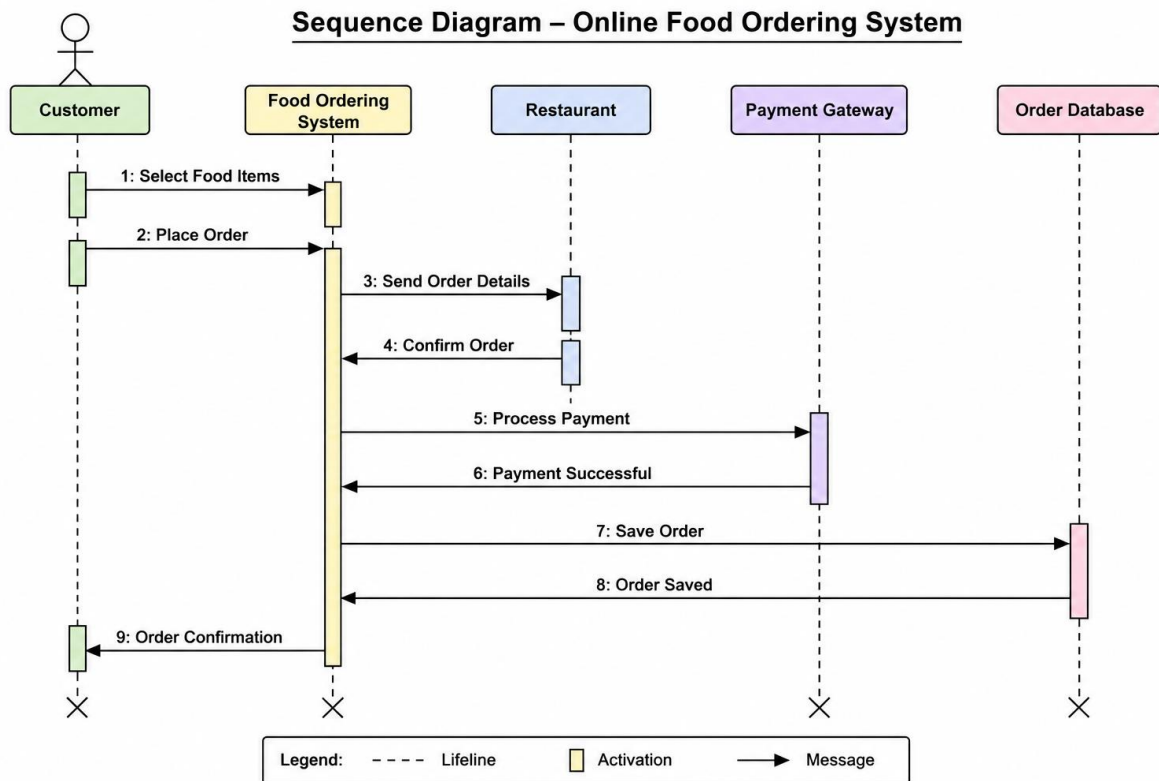
An activation bar represents the period during which an object performs an operation.

Examples:

- Food Ordering System is active while processing the order.
- Restaurant is active while confirming the order.

- Payment Gateway is active while processing the payment.
- Order Database is active while storing order details.

(d) Sequence Diagram



(10 Marks)

4. Activity Diagram Problem

A **Student Admission System** includes application submission, document verification, fee payment, and enrollment confirmation. The workflow lacks proper decision-making steps.

Tasks:

(a) Identify All Activities in the Workflow

The activities involved in the **Student Admission System** are:

1. Start
2. Submit Application
3. Verify Documents
4. Approve/Reject Documents
5. Pay Admission Fee
6. Verify Fee Payment
7. Generate Enrollment Number

8. Confirm Enrollment
9. End

(b) Add Decision Nodes for Approval/Rejection

The workflow includes the following decision points:

Decision 1: Documents Verified?

- **Yes** → Proceed to Fee Payment.
- **No** → Reject Application and End Process.

Decision 2: Fee Payment Successful?

- **Yes** → Generate Enrollment Number.
- **No** → Request Payment Again.

(c) Include Parallel Processing Where Applicable

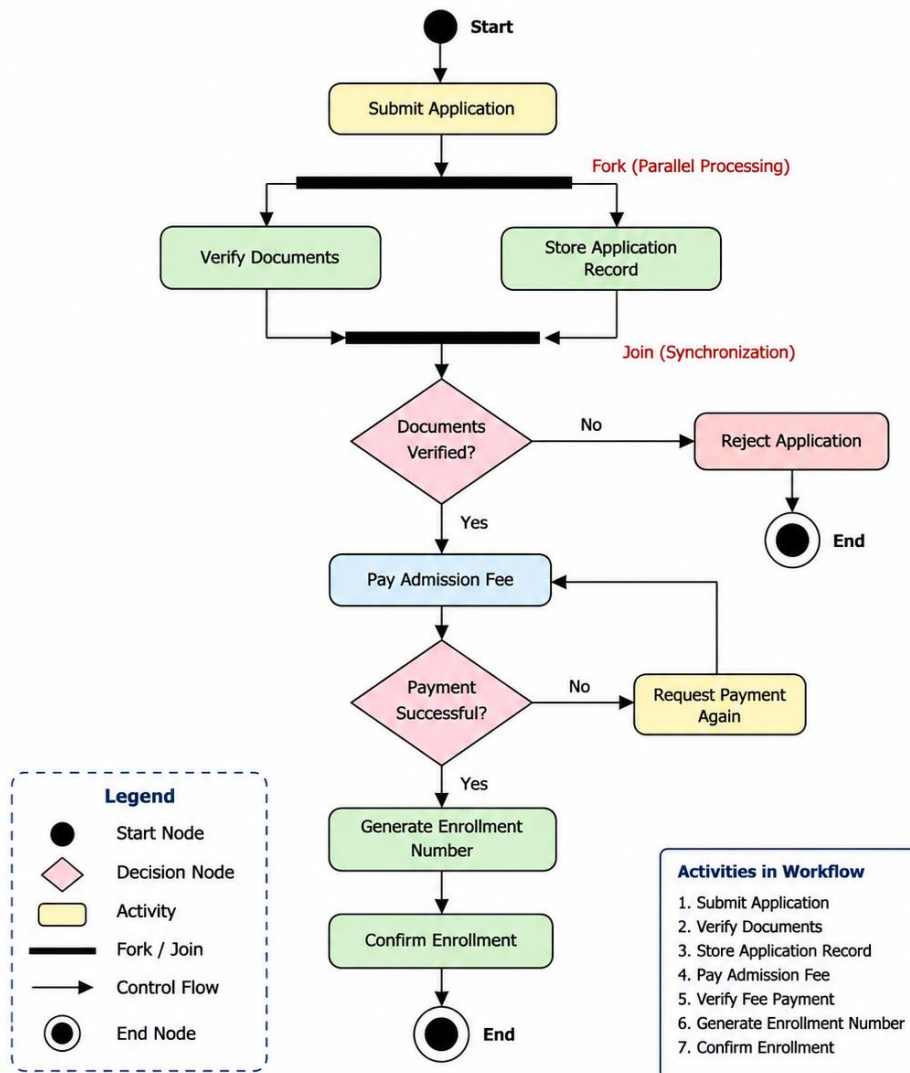
After the **application is submitted**, two activities can occur simultaneously:

- **Document Verification**
- **Application Record Storage**

These activities execute in parallel and synchronize before proceeding to the next step.

(d) Optimized Activity Diagram

Activity Diagram – Student Admission System



(10 Marks)

5. State Diagram Problem

A **Vending Machine System** operates through states such as Idle, Product Selected, Payment Received, Product Dispensed, and Transaction Completed.

Tasks:

(a) Identify All Possible States

The Vending Machine operates through the following states:

1. Initial State (Start)
2. Idle

3. **Product Selected**
4. **Payment Received**
5. **Product Dispensed**
6. **Transaction Completed**
7. **Final State (End)**

(b) Define Events Triggering Transitions

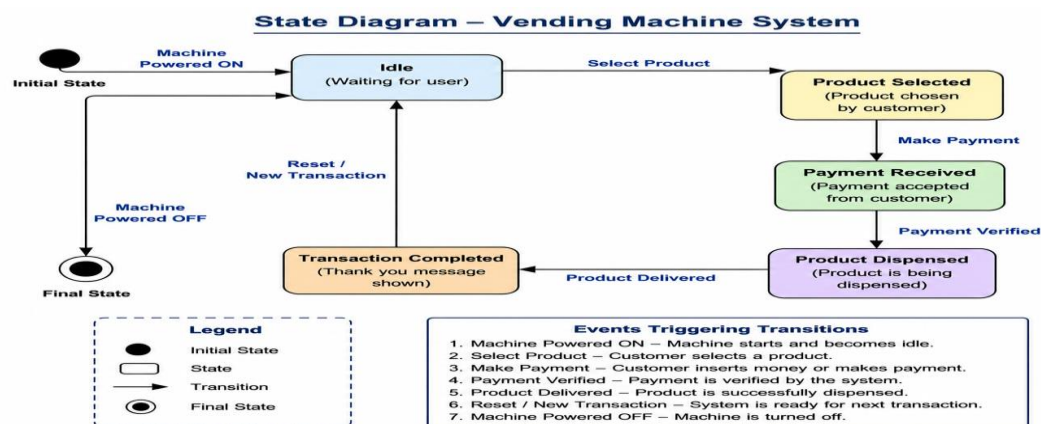
Current State	Event	Next State
Initial State	Machine Powered ON	Idle
Idle	Customer Selects Product	Product Selected
Product Selected	Customer Makes Payment	Payment Received
Payment Received	Payment Verified	Product Dispensed
Product Dispensed	Product Delivered	Transaction Completed
Transaction Completed	Transaction Ends / Reset	Idle
Idle	Machine Powered OFF	Final State

(c) Include Initial and Final States

- **Initial State:** Represented by a **filled black circle (●)**.
- **Final State:** Represented by a **double circle (◎)**.

The state transition begins from the **Initial State**, moves through all operational states, and ends at the **Final State** when the machine is switched off.

(d) State Transition Diagram



(10 Marks)

6. Component Diagram Problem

An **Online Learning Platform** consists of modules such as User Management, Course Management, Assessment Module, Content Delivery Module, and Notification Service.

Tasks:

(a) Identify Major System Components

The **Online Learning Platform** consists of the following major components:

1. **User Management**
2. **Course Management**
3. **Assessment Module**
4. **Content Delivery Module**
5. **Notification Service**
6. **Database**

Functions of Each Component

Component	Function
User Management	Manages user registration, login, and profiles.
Course Management	Handles course creation, updates, and enrollment.
Assessment Module	Conducts quizzes, assignments, and grading.
Content Delivery Module	Delivers videos, notes, and learning materials.
Notification Service	Sends alerts, emails, and announcements.
Database	Stores users, courses, assessments, and results.

(b) Define Interfaces Between Components

The interfaces describe how the components communicate.

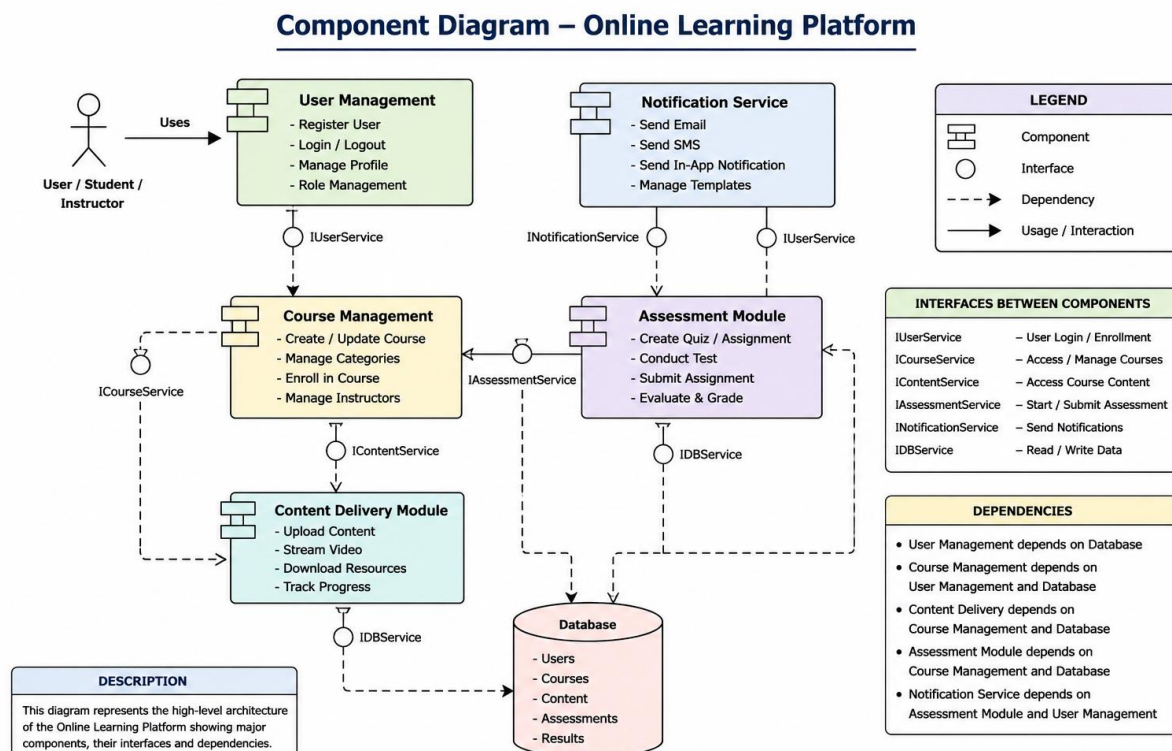
- **User Management ↔ Course Management**
 - User Login
 - Course Enrollment
- **Course Management ↔ Content Delivery Module**
 - Access Course Content
- **Course Management ↔ Assessment Module**
 - Start Quiz
 - Submit Assignment
- **Assessment Module ↔ Notification Service**
 - Send Result Notifications
- **All Components ↔ Database**
 - Read and Write Data

(c) Show Dependencies Among Modules

The dependencies among modules are:

- **User Management** depends on the **Database**.
- **Course Management** depends on **User Management** and the **Database**.
- **Content Delivery Module** depends on **Course Management**.
- **Assessment Module** depends on **Course Management** and the **Database**.
- **Notification Service** depends on the **Assessment Module** and **User Management**.

(d) Component Diagram:



(10 Marks)

7. Deployment Diagram Problem

A **Smart Traffic Monitoring System** uses roadside sensors to collect vehicle data. The data is transmitted to a central cloud server where traffic authorities monitor congestion through a web dashboard. Alerts are generated during traffic violations and accidents.

Tasks:

(a) Identify All Hardware Nodes Involved in the System

The Smart Traffic Monitoring System consists of the following hardware nodes:

1. **Roadside Sensor Node**
2. **Communication Network (Internet/4G/5G)**
3. **Cloud Server**
4. **Traffic Authority Computer (Web Dashboard)**
5. **Administrator/Officer Device**

(b) Specify Software Components Deployed on Each Node

Hardware Node	Software Components
Roadside Sensor Node	Vehicle Detection Software, Data Collection Module
Communication Network	Data Transmission Service
Cloud Server	Traffic Monitoring System, Alert Management System, Database
Traffic Authority Computer	Web Dashboard Application
Administrator Device	Alert Notification Application

(c) Define Communication Links Between Nodes

The communication between the nodes is as follows:

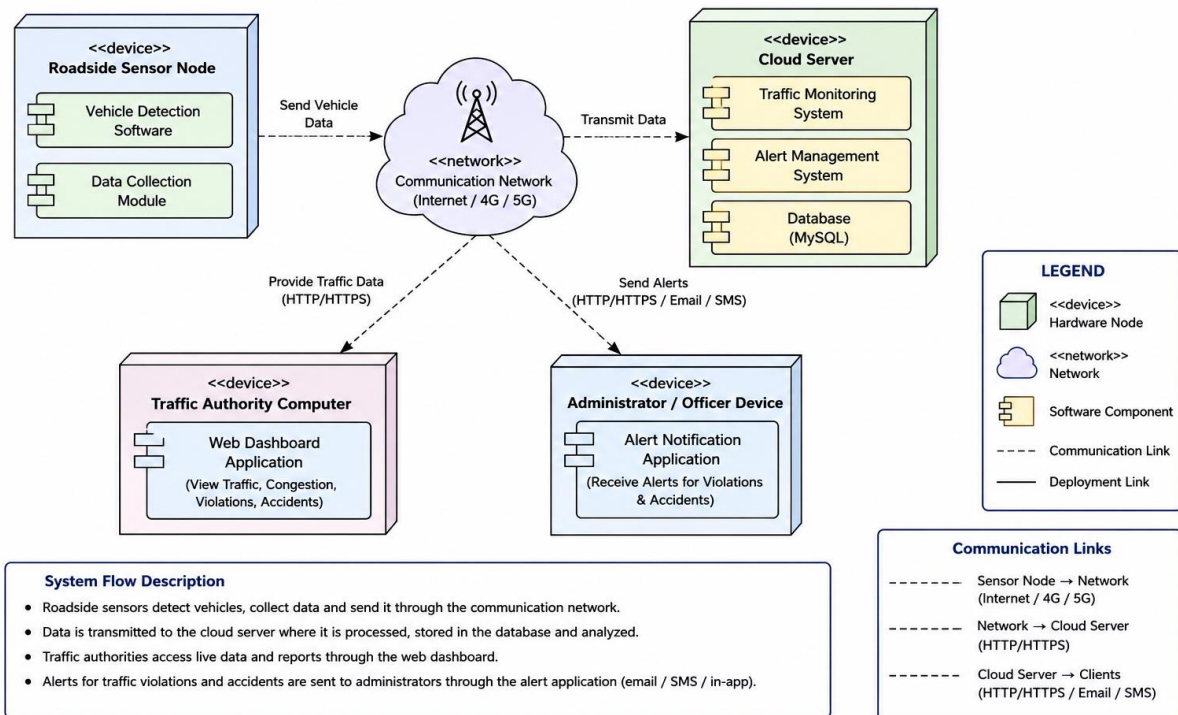
- **Roadside Sensor Node → Communication Network**
 - Sends vehicle and traffic data.
- **Communication Network → Cloud Server**
 - Transfers real-time traffic information.
- **Cloud Server → Traffic Authority Computer**
 - Displays live traffic status on the web dashboard.
- **Cloud Server → Administrator Device**
 - Sends alerts for traffic violations and accidents.

Communication can occur through:

- Internet
- Wi-Fi
- 4G/5G Network

(d) Deployment Diagram:

Deployment Diagram – Smart Traffic Monitoring System



(10 Marks)

RUBRICS FOR CALCULATION

ANALYTICAL PROBLEM SOLVING

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation